

Cover Letter: JMLR Machine Learning Open Source Software

kodama-cpp: Cross-Validated Accuracy Maximization on CPU, CUDA, and Apple Metal

Dear Editors of the Journal of Machine Learning Research,

We submit kodama-cpp for consideration in the Machine Learning Open Source Software track. The software is a standalone C++17 implementation of KODAMA with explicit CPU, NVIDIA CUDA, and Apple Metal backends and thin R and Python wrappers.

Submission identity

Open-source license: MIT project code with compatible retained third-party terms; see `PROVENANCE.md` and `THIRD_PARTY_NOTICES.md`

Project URL: <https://github.com/tkcaccia/kodama-cpp>

R wrapper URL: <https://github.com/tkcaccia/kodama-r>

Python binding source: <https://github.com/tkcaccia/kodama-cpp/tree/main/split-repos/kodama-python> (standalone repository publication pending)

Software version: 0.1.0 release candidate

Reviewed baseline: 9da48ee03d4ce76d50adf7cae0189b8ea7d65c62

Reviewed R-wrapper baseline: 4d5506bde8918162dae0ad1d19c4f6f7a4df64ba

Final release commit: AUTHOR ACTION REQUIRED: insert the full v0.1.0 tagged commit

Archived source: AUTHOR ACTION REQUIRED: deposit the tagged source archive, insert its URL and checksum, and add a DOI if assigned

Significance and software contribution

The submission preserves KODAMA's cross-validated label-predictability objective while moving its repeated work into one typed float32 core. It contributes package-owned nearest-neighbor, SIMPLS/LDA, graph, and visualization paths; strict and observable CPU/CUDA/Metal execution; matrix and supplied-graph entry points; and a common API for R and Python. The repository contains installation instructions, API examples, tests, benchmark drivers, a public API snapshot, and a machine-checkable licensing/provenance audit.

Prior publications and delta

The KODAMA method was published by Cacciatore, Luchinat, and Tenori in PNAS (2014), and the historical R package was described by Cacciatore et al. in Bioinformatics (2017). Prior publication of the method is disclosed. This submission concerns the new standalone software: a C++17 core independent of R, native CUDA and Metal backends, float32 and label-aware SIMPLS/LDA, package-owned neighbor search, graph-input APIs, strict backend metadata, and separately maintained R/Python bindings. The detailed delta is recorded in `docs/previous-version-delta.md`. The two prior papers will be supplied as supplementary material.

Evidence of use and openness

KODAMA has an established public predecessor: a peer-reviewed method paper, a peer-reviewed R-package paper, a maintained public R package, documentation, and public source history. The CRANlogs public API reports 48,480 KODAMA downloads from 2017-01-01 through 2026-07-26, including 4,526 during 2025-07-27--2026-07-26 (<https://cranlogs.r-pkg.org/downloads/total/2017-01-01:2026-07-26/KODAMA> and <https://cranlogs.r-pkg.org/downloads/total/2025-07-27:2026-07-26/KODAMA>). These counts concern the predecessor R package, not the new implementation. kodama-cpp itself is a new repository and, on 2026-07-27, had zero GitHub stars, forks, releases, and public issues; these figures are disclosed rather than presented as adoption. AUTHOR ACTION REQUIRED: refresh the dated counts and add any independently verifiable user or downstream-package evidence available immediately before submission. The repository provides an issue tracker, contribution guide, changelog, API-stability policy, reproducible tests, and release checklist.

Required author declarations

- Coauthor consent: AUTHOR ACTION REQUIRED. Confirm that all seven authors know of and consent to this submission.
- Funding supporting this work during the previous 36 months: AUTHOR ACTION REQUIRED.
- Competing interests and recent collaborations with JMLR action editors: AUTHOR ACTION REQUIRED.
- Suggested action editors: AUTHOR ACTION REQUIRED. Supply 3-5 conflict-free JMLR action editors with brief relevance statements.
- Suggested reviewers: AUTHOR ACTION REQUIRED. Supply 3-5 conflict-free reviewers with brief relevance statements.

Keywords

KODAMA; cross-validation; unsupervised learning; heterogeneous computing; manifold learning

Sincerely,

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